

Week 4: Project Summary

By Osuji Onyinyechi

HealthConnect Clinic Experience Lab

Data Science Track Problem Understanding & ML Problem Definition

Programme	AnalystLab Africa Experience Lab
Week	Week 4
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Track	Data Science

1. The Specific Problem This Track Will Address

The Data Science track is responsible for defining the machine learning problem underlying HealthConnect Clinic's patient no-show challenge, and for assessing whether the available appointment data can support a reliable no-show prediction solution. This includes proposing a target variable, identifying candidate input features, and establishing an initial modelling approach that later weeks will build upon.

2. Resources Used

- HealthConnect_Appointment_Data.csv, 5,000 anonymised appointment records across 18 variables, covering patient demographics, appointment details, booking information, appointment history, reminder information, distance to clinic, and appointment outcome.
- HealthConnect_Data_Dictionary, variable definitions, data types, and notes used to interpret the dataset correctly before analysis.

3. Key Observations from Week 4

- The dataset is clean and well-structured, with zero duplicate records and zero duplicate appointment identifiers.
- Missingness is limited and largely explainable: the 1,366 missing values in reminder_channel occur exactly where no reminder was sent, and distance_to_clinic_km (90 rows) and waiting_time_minutes (60 rows) each have under 2 percent missingness.
- The target variable, appointment_outcome, is close to evenly split between No-Show (48.5 percent) and Attended (46.3 percent), with Cancelled forming a small minority class (5.3 percent), which is favourable for building a balanced binary classifier.
- waiting_time_minutes was identified as a likely data leakage risk, since it is probably only recorded once a patient has already arrived, and would not be genuinely available at prediction time.
- The dataset contains 1,696 unique patients across 5,000 appointments, meaning many patients appear more than once. This has direct implications for how training and test data should be split.

4. Proposed Approach

The problem is framed as a binary classification task: predicting whether a scheduled appointment will result in a no-show. Cancelled appointments are excluded from the modelling dataset, since a cancellation reflects a proactive, communicated decision by the patient, while a no-show is a silent absence and the actual operational cost the clinic wants to reduce. This leaves 4,737 appointments for modelling, with a well-balanced target distribution of approximately 51 percent No-Show and 49 percent Attended.

Candidate input features span patient demographics, appointment characteristics, booking lead time, prior appointment and no-show history, and reminder information. An engineered prior no-show rate is proposed alongside the raw history counts. The initial modelling plan is to establish an interpretable baseline (logistic regression) before considering tree-based models, using a stratified, patient-aware train/test split, and prioritising precision and recall for the No-Show class over raw accuracy.

5. Key Considerations That May Affect the Project

- Data leakage: `waiting_time_minutes` must be excluded or clarified before use, since it likely postdates the outcome it would be used to predict.
- Patient-level leakage: because many patients have multiple appointments, a random row-level split risks letting the same patient appear in both training and test sets, inflating apparent performance.
- Synthetic data: the dataset is fictional and anonymised, so learned patterns may not fully generalise to real clinic behaviour.
- Scope boundary: Cancelled appointments are not modelled in this phase, so the resulting solution addresses no-shows specifically, not cancellations.
- Fairness: demographic variables such as gender and age are included as candidate features and will need review to ensure any resulting model does not produce inequitable outcomes across patient groups.

6. Proposed Focus for Week 5

Week 5 will begin exploratory data analysis on the finalised modelling dataset (No-Show versus Attended only), examining how the proposed features relate to the target variable. This includes confirming the treatment of `waiting_time_minutes`, validating the value of the engineered prior no-show rate feature, and building an initial baseline model to establish a performance benchmark ahead of further refinement.